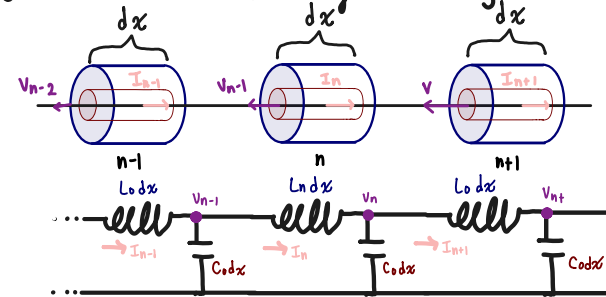


Review:

consider cables as many small segments

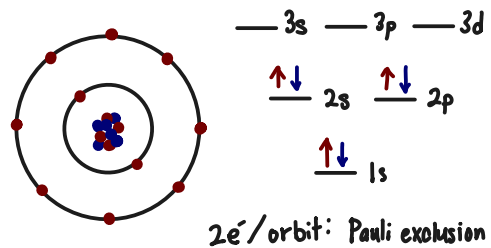


$$L_0 = \frac{\text{Inductance}}{\text{unit length}} \quad C_0 = \frac{\text{Capacitance}}{\text{unit length}}$$

$$C = \frac{1}{\sqrt{L_0 C_0}} = \text{speed}$$

$$Z_0 = \sqrt{\frac{L_0}{C_0}}$$

Energy Bands:



when atoms form a solid: $w \sim 10^{23}$ atoms $\sim 10^{23}$ states

* some electrons tightly bound to single ion core

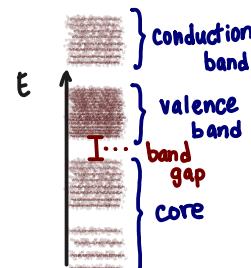
* some participate in bonds to hold solid together valence

* some are delocalized - travel throughout the crystal conduction

$$\frac{\tilde{V}_R}{\tilde{V}_0} = \frac{Z_L - Z_0}{Z_L + Z_0}$$

Reflected voltage

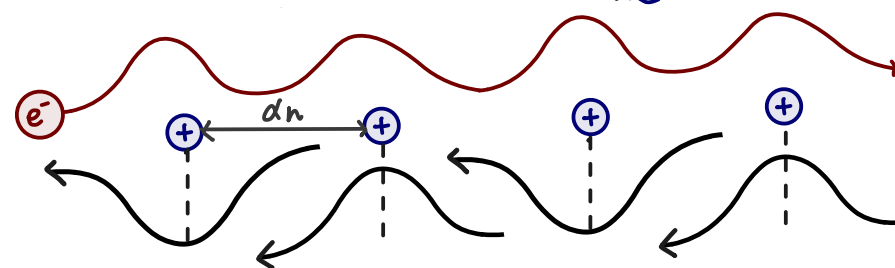
Input voltage



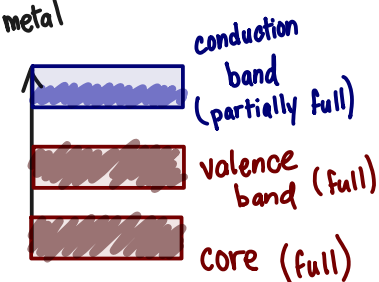
why band gaps? A crude hint of an explanation

consider e^- trying to move throughout a crystal

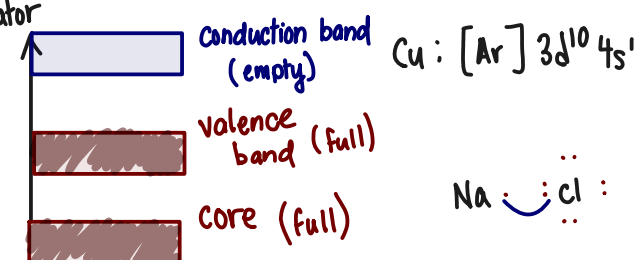
e^- leak like waves w/ deBroglie wavelength $\lambda = \frac{h}{p}$



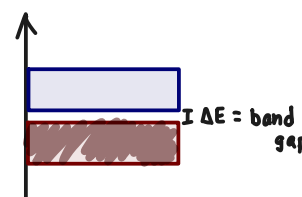
ex.) metal



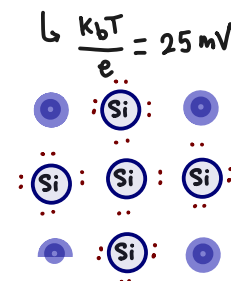
insulator



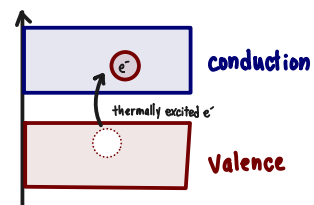
semiconductor



ΔE so small that $\Delta E \approx k_B T$

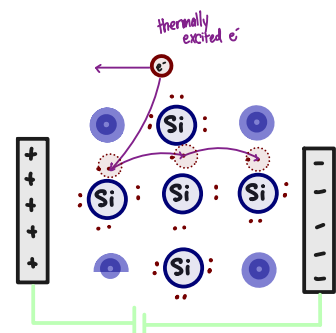


Semiconductor Holes:



hole: Vacancy in a valence band

that behaves much like a positively charged e^- .



hole looks like its moving towards negative electrode

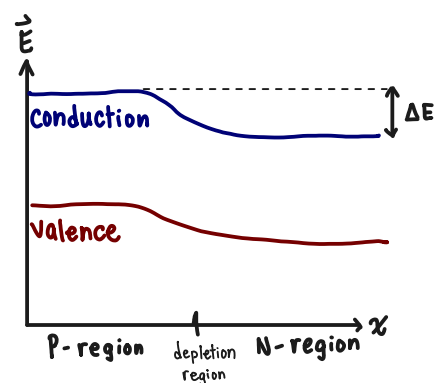
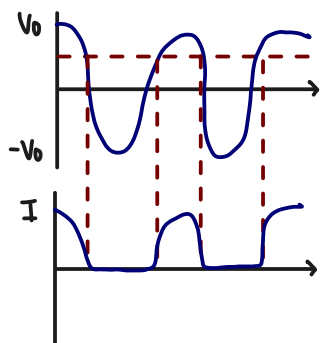
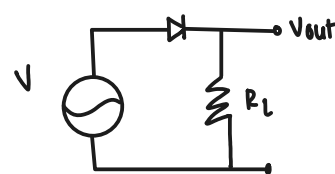
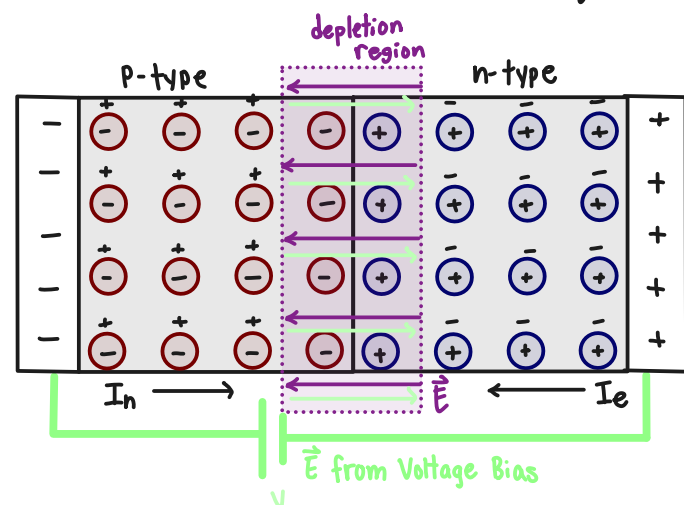
Doped Semiconductors

* atoms like P, Sb, As have 5 valence e^-

Doping: Replace Si w/ another type of atom

Doping with P, As; creates extra electronic states w/ e^- that can easily go into conduction band

P-type: doped, N-type: anti-doped



for thermal processes that involve hopping up a potential hill

$$I \propto e^{-\Delta E/kT}$$

$$\text{If we apply a bias } V$$

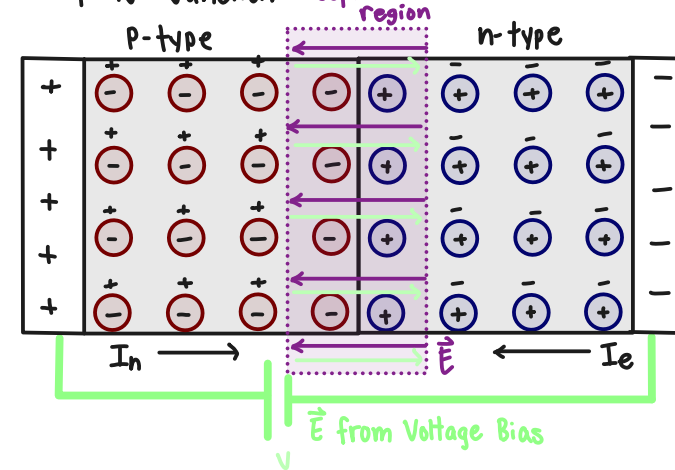
$$I \propto e^{-\frac{\Delta E - eV}{kT}} = e^{-\frac{\Delta E}{kT}} e^{eV/kT}$$

There is another small current for intrinsic carriers

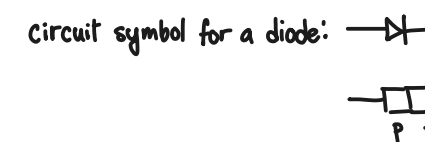
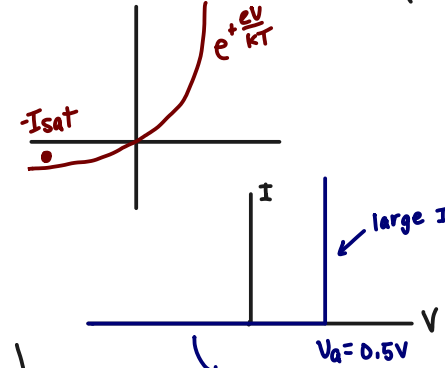
(thermally excited e^- in p-side + thermally excited holes in n-side)

$$I_{\text{sat}} \propto e^{-E_g/kT} \text{ @ } V=0 \text{ currents from directions cancel}$$

P-N Junction:



$$\text{It turns out: } I(V) = I_{\text{sat}}(e^{eV/kT} - 1)$$



Both V and n start out net neutral

* near the junction, some e^- from

side will diffuse over to p-side,

find a hole and recombine w/ e^-

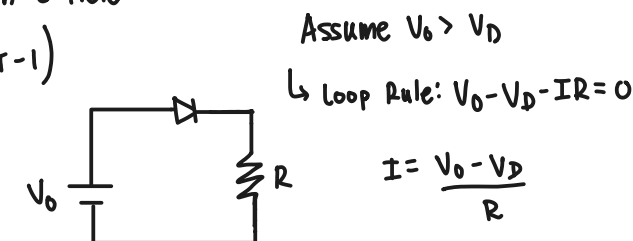
* near the junction we form

* depletion region w/ no mobile charge carriers

↳ Leaving ion cores and creating an E-field

* If we apply a voltage bias across

we can make current if $|E_{\text{bias}}| > |E_{\text{dep}}|$



Assume $V_0 > V_D$

↳ Loop Rule: $V_0 - V_D - IR = 0$

$$I = \frac{V_0 - V_D}{R}$$

Assume $V_0 < V_D$

$$I_D = 0 \Rightarrow I_R = 0$$

